COMPOSITIONAL SYNTHETIC CONTROLS

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Abstract. Many policy evaluations involve vectors of category-specific quantities, arising either from the aggregation of categorical individual-level data (e.g., employment type, major choice) or from compositional measures defined at the aggregate level (e.g., GDP by sector, votes by party, electricity generation by source). In such settings, the shares of each category and the total across categories can each carry independent economic meaning. Practitioners typically apply the Synthetic Control Method to a single category of interest or to each category separately. The first approach discards information about the remaining categories and cannot identify the source of any compositional shift. The second produces a different set of donor weights for each category, so the implied counterfactuals do not correspond to a single hypothetical donor unit: This paper proposes *compositional synthetic controls*, a method that constructs a single synthetic counterfactual vector using one unified set of weights obtained by minimizing a distance between log-transformed quantities. Under a standard random utility framework, these weights admit a behavioral interpretation as measuring the similarity between the treated unit's representative agent and those of the control units in terms of preferences over alternatives. Geometrically, the implied synthetic shares coincide with the Fréchet barycenter on the probability simplex equipped with the Aitchison metric. I illustrate the method by studying Colorado RPS policy on the electricity mix.

Keywords: categorical outcomes, compositional data, synthetic controls, simplex, Aitchison geometry, random utility model.

JEL codes: C01, C14, C4.

1. Introduction

Many empirical questions involve compositional outcomes: vectors of category-specific quantities that sum to a well-defined total. Such outcomes arise naturally when aggregating categorical individual-level data, for example, classifying workers by employment type (full-time, part-time, self-employed) or students by major (STEM, humanities, social sciences). They also arise at the aggregate level when tracking GDP by sector, votes by party, or electricity production by energy source. In each case, the full picture requires understanding both the relative distribution across categories (shares) and the overall scale (total), since both margins can carry independent economic meaning.

The Synthetic Control Method (SCM) provides a natural framework for causal inference in these settings, particularly when a single unit is exposed to an intervention at a known point in time and many unaffected units are available but no single one constitutes a valid counterfactual on its own. The classical formulation of Abadie and Gardeazabal (2003) and Abadie et al. (2010) constructs a weighted combination of control units that closely matches the treated unit's pre-intervention trajectory and then uses that combination as a proxy for the counterfactual post-intervention path.

When the outcome is compositional, practitioners commonly apply SCM to a single category of interest or to each category separately. Neither approach is fully satisfactory. Focusing on a single category discards information about the remaining components and cannot identify the source of any compositional shift. For example, in studying electricity generation by source, it is silent on whether a rise in the renewable quantity of electricity generation reflects a reduction in fossil-fuel output, a collapse of nuclear, or simply faster growth in total generation. Applying SCM separately to each category avoids this omission but creates a different problem: the donor weights differ across categories, so the implied counterfactuals do not correspond to a single hypothetical unit. The synthetic fossil-fuel state and the synthetic renewable state are different weighted averages of the control pool, and any comparison between them mixes incompatible counterfactuals. As noted by Tian et al. (2023) and Sun et al. (2025), the shared latent structure underlying the components provides a natural rationale for using instead a single unified set of weights that leverages that common information.

Beyond the inconsistency of weights, two further difficulties arise. First, matching on raw quantities when units differ substantially in size can distort the weighting procedure (Abadie, 2021): a large state and a small state with identical energy mix compositions will appear far apart in levels even though their relative structures are the same, pushing the optimizer toward scale matches rather than structural ones. Second, when the analysis focuses on shares rather than levels, the unit-sum constraint means that what is economically relevant are ratios and relative proportions, not raw numerical differences. The Euclidean distance underlying classical SCM treats each component independently and therefore fails to respect the intrinsic geometry of the simplex: a shift from 1% to 2% renewable generation is economically very different from a shift from 50% to 51%, yet both register as a one-percentage-point change.

The distortion introduced by Euclidean matching is easy to illustrate. Consider three regions (r_1, r_2, r_3) each allocating GDP across services, industry, and agriculture. Region r_1 allocates $(82.5\%, 15.0\%, 0.5\%)$; region r_2 allocates $(72.5\%, 25.0\%, 0.5\%)$; region r_3 allocates $(92.5\%, 5.0\%, 0.5\%)$. The Euclidean distance from r_1 to each of r_2 and r_3 equals 0.14, making them appear identical matches. Yet in r_1 the services-to-industry ratio is 5.5, while in r_2 it is 2.9 and in r_3 it is 18.5. Region r_2 thus exhibits a far closer economic structure to r_1 than r_3 does. Euclidean distance treats them symmetrically and therefore produces misleading synthetic controls.

This paper develops *compositional synthetic controls* (CSC), a method that constructs a single synthetic counterfactual vector of quantities from which both the synthetic totals and shares can be derived, using one set of economically meaningful weights. The weights are chosen to minimize the squared Euclidean distance between the log-transformed quantity vectors of the treated unit and the weighted combination of control units during the pre-intervention period. Log transformation addresses scale differences and, as shown below, imposes a structure that is directly interpretable in terms of relative proportions.

The paper makes three main contributions. First, it provides a behavioral foundation by embedding the estimator within a random utility framework (McFadden, 1974). Under this interpretation, the synthetic counterfactual corresponds to a hypothetical representative agent whose expected utilities are a weighted average of those of the control agents. The

weights, therefore, capture how closely each control unit's preference structure approximates that of the treated unit.

Second, it provides a geometric characterization. The simplex equipped with the Aitchison metric (Aitchison, 1982; Egozcue et al., 2003) is the natural ambient space for compositional data, where distances and averages are governed by log-ratio relationships. I show that the synthetic shares constructed by CSC coincide with the Fréchet barycenter of the control units' share vectors under this geometry, ensuring coherence with the compositional structure.

Third, the paper proposes a placebo-based inference procedure that extends the approach of Abadie and Gardeazabal (2003) to the multivariate compositional setting, providing a practical and transparent tool for assessing the significance of estimated effects.

Related literature. This paper contributes to several bodies of work. On extensions of the synthetic control method, Doudchenko and Imbens (2016) and Abadie and L'Hour (2017) propose flexible balancing approaches that relaxes the non-negativity and adding-up constraints of the classical method. Amjad et al. (2018) and Arkhangelsky et al. (2019) improve robustness and inference under model uncertainty. Athey et al. (2018), Ben-Michael et al. (2018), Chernozhukov et al. (2018), and Imai and Kim (2019) explore alternative designs, penalization schemes, and inferential strategies for high-dimensional or noisy settings. Viviano and Bradic (2019) develops design-based inference. More recently, Gunsilius (2023) and Gunsilius et al. (2024) generalize synthetic controls to distributional outcomes, paving the way for applications where the object of interest is a probability distribution rather than a scalar.

On multivariate outcomes, Tian et al. (2023) and Sun et al. (2025) discusses the advantages of using a common set of weights when outcomes share latent factors. The present paper provides a formal justification for this in the compositional setting through both a structural model and a geometric argument.

On causal inference for non-Euclidean data, the closest antecedents are Zhu and Müller (2023), Kurisu et al. (2024), and Boussim (2025). The most directly related work is Kurisu et al. (2025), which develops synthetic controls for general manifold-valued outcomes. The present paper differs in focus: rather than targeting abstract metric spaces, it provides a

specialized, economically interpretable framework for compositional outcomes in which the geometry has explicit behavioral content.

The paper is organized as follows. Section 2 presents the analytical framework and defines the treatment effects of interest. Section 3 describes the CSC estimator, the smearing correction for retransformation bias, and the placebo test. Section 4 develops the random utility interpretation of the weights. Section 5 provides the geometric characterization in terms of Fréchet barycenters. Section 7 concludes. Proofs are collected in the Appendix.

2. Analytical Framework

2.1. **Setup and notation.** Consider J units indexed by $j = 1, \ldots, J$ observed over T periods indexed by $t = 1, \ldots, T$. For each unit j and period t, the outcome is a p-dimensional vector of category-specific quantities:

$$q_{j,t} = \big(q_{1,j,t},\, q_{2,j,t},\, \ldots,\, q_{p,j,t}\big) \in \mathbb{R}^p_{>0}.$$

The total quantity for unit j at time t is

$$S_{j,t} = \sum_{k=1}^{p} q_{k,j,t},$$

and the corresponding vector of category shares is

$$\pi_{k,j,t} = \frac{q_{k,j,t}}{S_{j,t}}, \qquad k = 1, \ldots, p,$$

so that $\pi_{j,t} = (\pi_{1,j,t}, \ldots, \pi_{p,j,t})$ lies in the open probability simplex $\mathcal{S}^{p-1} = \{v \in \mathbb{R}^p_{>0} : \sum_k v_k = 1\}$. An intervention occurs at a known time T_0. Unit $j = 1$ is the sole treated unit; units $j = 2, \ldots, J$ form the donor pool. Using potential outcomes notation, let $q^0_{j,t}$ denote the outcome in the absence of treatment and $q^1_{j,t}$ the outcome under treatment. The standard no-interference assumption requires that the intervention does not affect the control units:

$$q^0_{j,t} = q^1_{j,t}, \qquad \forall\, j \neq 1, \quad \forall\, t.$$

For the treated unit, the pre-intervention outcomes are unaffected by the intervention:

$$q^0_{1,t} = q^1_{1,t}, \qquad \forall\, t \leq T_0,$$

while in the post-intervention period the observed and counterfactual outcomes generally differ. Only $q^1_{1,t}$ is observed for $t > T_0$; the counterfactual $q^0_{1,t}$ is the primary object of inference.

2.2. **Treatment effects of interest.** Once the counterfactual vector $q^0_{1,t}$ is constructed, three complementary treatment effects can be defined.

Category-specific effects. For each category k, the treatment effect on the quantity level is

$$\mathrm{CTE}_{k,t} = \frac{q^1_{k,1,t}}{q^0_{k,1,t}} - 1,$$

where $q^1_{k,1,t}$ is the observed post-treatment quantity and $q^0_{k,1,t}$ is its counterfactual.

Effect on the total. Let $S^1_{1,t} = \sum_k q^1_{k,1,t}$ and $S^0_{1,t} = \sum_k q^0_{k,1,t}$. The treatment effect on the aggregate is

$$\mathrm{TTE}_t = \frac{S^1_{1,t}}{S^0_{1,t}} - 1.$$

Effect on shares. Denoting by $\pi^1_{k,1,t} = q^1_{k,1,t}/S^1_{1,t}$ and $\pi^0_{k,1,t} = q^0_{k,1,t}/S^0_{1,t}$ the observed and counterfactual shares respectively, the share treatment effect is

$$\mathrm{STE}_{k,t} = \pi^1_{k,1,t} - \pi^0_{k,1,t}.$$

These three parameters provide a unified decomposition of the treatment effect: $\mathrm{CTE}_{k,t}$ captures the impact on each category in levels; TTE_t captures the effect on the overall scale; and $\mathrm{STE}_{k,t}$ captures the reallocation across categories holding the total fixed. Note that constructing a single counterfactual vector $q^0_{1,t}$ is sufficient to recover all three, ensuring internal consistency: the synthetic shares and totals are derived from the same underlying weights.

3. The Compositional Synthetic Control Estimator

3.1. **Motivating data model.** Suppose the untreated potential outcome for category k, unit j, at time t satisfies the interactive fixed-effects model for log-quantities:

$$\log q^0_{k,j,t} = \delta_{k,t} + Z_j'\theta_{k,t} + \mu_j'\lambda_{k,t} + \varepsilon_{k,j,t}, \tag{1}$$

where $\delta_{k,t}$ is a category-time fixed effect, Z_j and μ_j are $r \times 1$ and $m \times 1$ vectors of observed and unobserved unit-level characteristics with category-specific loadings $\theta_{k,t}$ and $\lambda_{k,t}$, and $\varepsilon_{k,j,t}$ is a transitory shock. Model (1) closely parallels those in Abadie et al. (2010) and Abadie (2021), with the key difference that it operates on log-quantities and permits the loadings to vary across categories.

The critical assumption is that the unit-specific unobserved predictor μ_j is *common* to all categories within the same unit. This shared factor structure is the economic rationale for seeking a single set of weights rather than separate synthetic controls per category. As Sun et al. (2025) notes, category-specific unobserved predictors can be accommodated provided they are orthogonal to the common component.

Working with log-transformed quantities has two further advantages. First, it addresses scale disparities: states or regions that differ dramatically in population size are rendered more comparable on the log scale. Second, as shown in Sections 4 and 5, the log transformation is the natural one from both the random utility and the Aitchison geometry perspectives.

3.2. **Weight estimation.** Let $\lambda = (\lambda_2, \ldots, \lambda_J)'$ denote the vector of weights assigned to the control units. The feasible set of weights is

$$\Lambda_{J-1} = \Big\{ v \in \mathbb{R}^{J-1} : v_j \geq 0 \ \forall\, j,\ \textstyle\sum_{j=2}^{J} v_j = 1 \Big\}.$$

The optimal weights are

$$\lambda^* = \arg\min_{\lambda \in \Lambda_{J-1}} \sum_{t=1}^{T_0} \Big\| \log q_{1,t} - \textstyle\sum_{j=2}^{J} \lambda_j \log q_{j,t} \Big\|^2, \tag{2}$$

where $\log q_{j,t} = (\log q_{1,j,t}, \ldots, \log q_{p,j,t})$ and $\|\cdot\|$ is the Euclidean norm on $\mathbb{R}^p$. This is a constrained least-squares problem and can be solved efficiently via standard quadratic programming routines. The counterfactual quantity for category k is:

$$\hat{q}^0_{k,1,t} = \exp\Big(\sum_{j=2}^{J} \lambda^*_j \log q_{k,j,t} \Big) = \prod_{j=2}^{J} q_{k,j,t}^{\lambda^*_j}, \tag{3}$$

which is the weighted geometric mean with weights λ^*. The synthetic total and synthetic shares follow:

$$S^0_{1,t} = \sum_{k=1}^{p} \hat{q}^0_{k,1,t}, \qquad \pi^0_{k,1,t} = \hat{q}^0_{k,1,t} / S^0_{1,t}. \tag{4}$$

3.3. **Inference via placebo tests.** To assess the statistical credibility of the estimated effects, I implement a placebo permutation test following Abadie and Gardeazabal (2003). The procedure re-applies the CSC estimator to each control unit $j = 2, \ldots, J$, treating it as if it had received the intervention at time T_0 and using the remaining $J-2$ control units as its donor pool. This yields a placebo synthetic control $q^{\mathrm{sc}(j)}_{j,t}$ for each j.

For any unit j, the pre-treatment and post-treatment root mean squared prediction errors (RMSPE) are:

$$\mathrm{RMSPE}^{(j)}_{\mathrm{pre}} = \Big[\frac{1}{T_0}\sum_{t=1}^{T_0}\big\|q_{j,t} - q^{\mathrm{sc}(j)}_{j,t}\big\|^2\Big]^{1/2}, \qquad \mathrm{RMSPE}^{(j)}_{\mathrm{post}} = \Big[\frac{1}{T-T_0}\sum_{t=T_0+1}^{T}\big\|q_{j,t} - q^{\mathrm{sc}(j)}_{j,t}\big\|^2\Big]^{1/2}.$$

To guard against unreliable placebo controls that fit the pre-period poorly, I follow the recommendation of Abadie et al. (2010) and discard placebo units for which $\mathrm{RMSPE}^{(j)}_{\mathrm{pre}}$ substantially exceeds that of the treated unit. The test statistic for unit j is the post-to-pre RMSPE ratio:

$$R^{(j)} = \frac{\mathrm{RMSPE}^{(j)}_{\mathrm{post}}}{\mathrm{RMSPE}^{(j)}_{\mathrm{pre}}}.$$

A large value of $R^{(1)}$ relative to the placebo distribution $\{R^{(j)}\}_{j=2}^{J}$ provides evidence of a treatment effect. The exact one-sided p-value is

$$p = \frac{1 + \sum_{j=2}^{J}\mathbb{I}\big(R^{(j)} \geq R^{(1)}\big)}{J}.$$

Under the null hypothesis of no treatment effect, the treated unit is exchangeable with the controls, so the treated unit's ratio should not systematically exceed those of the placebos.

Remark 1 (Multivariate RMSPE)**.** *The RMSPE above is computed on the full p-dimensional quantity vector. Practitioners may prefer to report separate RMSPEs for the total $S_{j,t}$ and for the share vector $\pi_{j,t}$, providing a more granular picture of where the post-treatment divergence occurs.*

4. Economic Interpretation of the Weights

I now provide a behavioral foundation for the CSC estimator by embedding it in a Random Utility Model (RUM). This interpretation clarifies the economic content of the weights λ^* and justifies the log transformation as the natural metric for compositional outcomes.

4.1. The random utility framework.

Treat each unit j as a *representative agent* who selects one option from a finite set of mutually exclusive alternatives $\{1, \ldots, p\}$. In the absence of treatment, the latent utility associated with alternative k for unit j at time t is

$$U^0_{k,j,t} = V^0_{k,j,t} + \varepsilon^0_{k,j,t}, \tag{5}$$

where $V^0_{k,j,t}$ is the systematic component and $\varepsilon^0_{k,j,t}$ is an idiosyncratic shock. The unit selects the alternative with the highest utility.

To separate the forces governing shares from those determining the total scale, decompose the systematic utility as

$$V^0_{k,j,t} = \mu^0_{k,j,t} - \log\Big(\sum_{\ell=1}^{p} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t}, \tag{6}$$

where $\mu^0_{k,j,t}$ captures the relative attractiveness of alternative k (governing the composition of choices) and the remaining terms ensure that the scale of choices is consistent with the observed total $S^0_{j,t}$.

Assume the shocks $(\varepsilon^0_{1,j,t}, \ldots, \varepsilon^0_{p,j,t})$ are independent and follow standard Type-I extreme value (Gumbel) distributions. The choice probabilities then take the multinomial logit form, and the predicted quantity for category k is

$$q^0_{k,j,t} = \pi^0_{k,j,t} \cdot S^0_{j,t} = \frac{e^{\mu^0_{k,j,t}}}{\sum_\ell e^{\mu^0_{\ell,j,t}}} \cdot S^0_{j,t}. \tag{7}$$

Taking logarithms of (7) yields

$$\log q^0_{k,j,t} = \mu^0_{k,j,t} - \log\Big(\sum_{\ell} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t}. \tag{8}$$

The expected utility of alternative k for unit j (using the well-known formula for the expectation of the maximum of Gumbel variates) is

$$\mathbb{E}[U^0_{k,j,t}] = \mu^0_{k,j,t} - \log\Big(\sum_{\ell} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t} + \gamma, \tag{9}$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. Comparing (8) and (9):

$$\log q^0_{k,j,t} = \mathbb{E}[U^0_{k,j,t}] - \gamma. \tag{10}$$

4.2. **Interpretation of the weights.**

Proposition 1 (Expected utility interpretation)**.** *Under the RUM described above, the counterfactual construction in equation (3) is equivalent to:*

$$\mathbb{E}\big[U^0_{1,t}\big] = \sum_{j=2}^{J} \lambda^*_j \, \mathbb{E}\big[U^0_{j,t}\big],$$

where $\mathbb{E}[U^0_{j,t}] = (\mathbb{E}[U^0_{1,j,t}], \ldots, \mathbb{E}[U^0_{p,j,t}])$.

Proof. See Appendix A. □

Proposition 1 states that the CSC counterfactual is equivalent to positing that the treated unit's representative agent, absent the intervention, would have had expected utilities equal to the weighted average of the control agents' expected utilities. The weights λ_j^* therefore measure the degree of preference similarity between the treated unit and each control unit: a higher weight on unit j indicates that its representative agent's choice behavior more closely reflects the latent preference structure of the treated unit. This interpretation provides both an intuitive rationale for the estimator and a transparent mapping from observed heterogeneity among control units to the treated unit's hypothetical preferences.

5. Geometric Interpretation

5.1. **The Aitchison geometry of the simplex.** Categorical distributions, such as category shares, reside in the probability simplex. In this space, the economically relevant object is not the absolute value of each component but its relationship to the others: each component represents a part of a fixed whole, and only ratios between components carry intrinsic information. Rescaling all components by a positive constant leaves the compositional structure unchanged.

Aitchison (1982) introduced a geometry tailored to this structure. The central object is the *Aitchison distance*, which measures dissimilarity through log-ratios:

$$\delta_A(\pi_1, \pi_2) = \sqrt{\frac{1}{2p} \sum_{i=1}^{p} \sum_{j=1}^{p} \left(\log \frac{\pi_{1,i}}{\pi_{1,j}} - \log \frac{\pi_{2,i}}{\pi_{2,j}} \right)^2}, \tag{11}$$

for $\pi_1, \pi_2 \in \mathcal{S}^{p-1}$. This metric is scale-invariant (dividing all components by the same positive constant leaves distances unchanged), ensures that a change from 1% to 2% is treated differently from a change from 50% to 51%, and satisfies the principle of subcompositional coherence, meaning that distances are preserved when a subset of categories is analyzed in isolation.

5.2. **Fréchet barycenters.** A fundamental challenge in non-Euclidean spaces is the definition of a meaningful notion of average. In Euclidean space, the weighted arithmetic mean minimizes the weighted sum of squared Euclidean distances. In a general metric space, this motivational property defines the Fréchet barycenter.

Definition 1 (Fréchet barycenter)**.** *Let (Mc, d) be a metric space, let $\{Y_j\}_{j=1}^N \subset Mc$, and let $\{w_j\}_{j=1}^N$ be non-negative weights summing to one. The Fréchet barycenter of $\{Y_j\}$ with weights $\{w_j\}$ is*

$$\bar{Y} = \arg\min_{m \in Mc} \sum_{j=1}^{N} w_j \, d^2(Y_j, m).$$

In Euclidean space with d the standard Euclidean distance, the Fréchet barycenter reduces to the weighted arithmetic mean. In the simplex with $d = \delta_A$ (the Aitchison distance), the Fréchet barycenter has an explicit closed form: it is the *closed geometric mean*, also known as the weighted geometric mean renormalized to the simplex.

Proposition 2 (Geometric characterization)**.** *The counterfactual shares in equation (4) coincide with the Fréchet barycenter of the control shares $\{\pi_{j,t}\}_{j=2}^J$ with weights $\{\lambda_j^*\}$ on the probability simplex $\mathcal{S}^{p-1}$ equipped with the Aitchison metric.*

Proof. See Appendix B. □

An alternative representation of the synthetic shares is obtained through the multinomial logit transform. Define $\ell : \mathcal{S}^{p-1} \to \mathbb{R}^{p-1}$ by

$$\ell(\pi) = \Big[\log \frac{\pi_1}{\pi_p}, \ldots, \log \frac{\pi_{p-1}}{\pi_p}\Big],$$

with inverse (softmax)

$$\ell^{-1}(y) = \frac{1}{1 + \sum_{i=1}^{p-1} e^{y_i}} \big(e^{y_1}, \ldots, e^{y_{p-1}}, 1\big)$$

for $y \in \mathbb{R}^{p-1}$. The choice of reference category p is arbitrary. Then the synthetic shares can be written as

$$\pi_{1,t}^0 = \ell^{-1}\Big(\sum_{j=2}^{J} \lambda_j^* \, \ell(\pi_{j,t})\Big).$$

This formulation makes explicit that the CSC estimator computes the Aitchison average by first mapping shares to log-odds (a bijection from the simplex to $\mathbb{R}^{p-1}$), computing the weighted arithmetic average in the unconstrained space, and then mapping back to the simplex via the softmax. The result is automatically a valid composition.

Remark 2 (Subcompositional coherence)**.** *Because the Aitchison metric satisfies subcompositional coherence, the CSC estimator automatically respects this property: if a subset of*

categories is analyzed in isolation, the estimated treatment effects on those categories are consistent with the effects obtained from the full compositional analysis.

Remark 3 (Comparison with separate synthetic controls)**.** *If instead one constructs separate synthetic controls for each share $\pi_{k,j,t}$, the resulting counterfactual shares will in general not sum to one and will be generated by p different sets of weights, producing p inconsistent synthetic treated units. The CSC estimator resolves both problems simultaneously by construction.*

6. Empirical Illustration: Colorado Renewable Portfolio Standard

6.1. **Policy background and data.** Colorado's Senate Bill 07-052, signed by Governor Bill Ritter in June 2007, expanded the state's Renewable Portfolio Standard from a 10% target to a mandate of 20% of retail electricity sales from eligible renewable sources by 2020. At the time of enactment, it was the most ambitious state-level renewable energy mandate in the United States, and it is widely regarded as a landmark in U.S. energy policy. A subsequent 2010 expansion raised the target to 30% for investor-owned utilities, and further amendments eventually set a 100% clean energy goal by 2040. Senate Bill 07-052 thus initiated a sustained structural shift in Colorado's electricity generation mix.

The data consist of annual electricity generation by source for all fifty U.S. states and the District of Columbia, covering 1990 to 2023, drawn from the U.S. Energy Information Administration. The compositional outcome is defined over two categories: (i) fossil-fuel-and-gas generation, denoted *fossil-and-gas* and equal to the sum of coal, oil, and natural gas generation; and (ii) renewable generation, comprising hydro, geothermal, wind, solar, and other renewables. The two-category specification is the natural one for evaluating an RPS, which directly targets the renewable share of total generation. The treatment date is $T_0 = 2006$, giving a pre-treatment period of 1990 to 2006 (17 years) and a post-treatment period of 2007 to 2023 (17 years).

6.2. **Donor pool construction.** The donor pool excludes states whose electricity mix may have been independently affected by comparable policies. In particular, I exclude the eleven RGGI states (Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Rhode Island, Vermont, and Virginia), which participated in a

cap-and-trade program for the power sector beginning in 2009. States with zero generation in either category in any sample year are also excluded, as the log transformation requires strictly positive values. After applying these criteria, 37 states remain: Alaska, Alabama, Arkansas, Arizona, California, Florida, Georgia, Hawaii, Iowa, Idaho, Illinois, Indiana, Kansas, Kentucky, Michigan, Minnesota, Missouri, Mississippi, Montana, North Carolina, North Dakota, Nebraska, New Mexico, Nevada, Ohio, Oklahoma, Oregon, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Washington, Wisconsin, West Virginia, and Wyoming.

6.3. **Pre-treatment fit.** The optimal weights are obtained by solving (2) over the 17 pre-treatment periods. Six states receive positive weight: Mississippi (0.336), Utah (0.327), New Mexico (0.110), Texas (0.082), California (0.075), and South Carolina (0.071). This combination reflects Colorado's pre-period generation profile, which is characterized by a high fossil-and-gas share (approximately 94%) and a modest renewable share (approximately 6%) driven primarily by hydroelectric generation.

Table 1 reports pre-treatment balance statistics. The log-level RMSPE over the pre-treatment period is 0.081, with fossil-and-gas at 0.035 and renewables at 0.109. Both values indicate an excellent pre-treatment fit. The share-based RMSPE of 0.005 confirms that the composition of Colorado's generation mix is essentially indistinguishable from the synthetic control during the pre-treatment period: the pre-period mean log-quantities for fossil-and-gas and renewables differ by less than 0.001 between Colorado and the synthetic control.

Figure 1 displays the log-quantity trajectories over the full sample. The synthetic control tracks Colorado with high precision throughout the pre-treatment period for both categories. After 2006, the two series diverge: Colorado's fossil-and-gas log-quantity falls progressively below the synthetic counterfactual, while its renewable log-quantity rises substantially above it, consistent with a compositional shift away from fossil-fuel generation and toward renewables.

6.4. **Estimated treatment effects.** Figure 2 shows the evolution of generation shares. Colorado's renewable share rises consistently above the synthetic counterfactual after 2007, with the fossil-and-gas share falling correspondingly. Both divergences widen monotonically, consistent with the progressive tightening of the RPS mandate.

Table 1. Pre-treatment balance: Colorado versus synthetic Colorado (1990–2006)

	Colorado	Synthetic Colorado
Mean log generation (MWh)		
Fossil-and-gas	17.461	17.460
Renewables	14.326	14.325
Donor weights		
Mississippi		0.336
Utah		0.327
New Mexico		0.110
Texas		0.082
California		0.075
South Carolina		0.071
Pre-treatment fit statistics		
Log-level RMSPE		0.081
Fossil-and-gas		0.035
Renewables		0.109
Share RMSPE		0.005

Note. Log-level RMSPE is the root mean squared prediction error of $\log q_{k,1,t} - \sum_j \lambda_j^* \log q_{k,j,t}$, pooled across the 17 pre-treatment periods $t = 1990, \ldots, 2006$ and both categories. Share RMSPE is the root mean squared prediction error of $\pi_{k,1,t} - \hat{\pi}^0_{k,1,t}$, pooled across periods and categories.

Table 2 reports the estimated treatment effects. The renewable share treatment effect ($\mathrm{STE}_{\mathrm{ren}}$) is positive in every post-treatment year, growing monotonically from 1.9 percentage points in 2007 to 23.9 percentage points in 2022. The category-level treatment effect on renewables ($\mathrm{CTE}_{\mathrm{ren}}$) is uniformly positive and large, reflecting that Colorado's absolute renewable generation grew far beyond the counterfactual prediction. The fossil-and-gas category treatment effect is uniformly negative, confirming that fossil-fuel generation was reduced

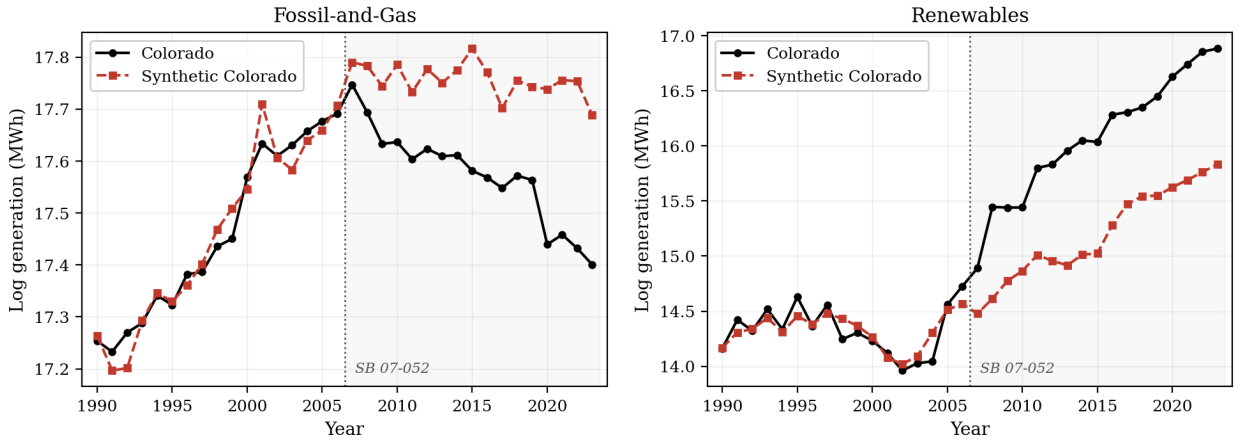

Figure 1. Log generation by category: Colorado versus synthetic Colorado, 1990–2023. *Note.* The vertical dotted line marks the treatment year (2006). The shaded area indicates the post-treatment period. The synthetic control is a weighted combination of Mississippi (0.336), Utah (0.327), New Mexico (0.110), Texas (0.082), California (0.075), and South Carolina (0.071).

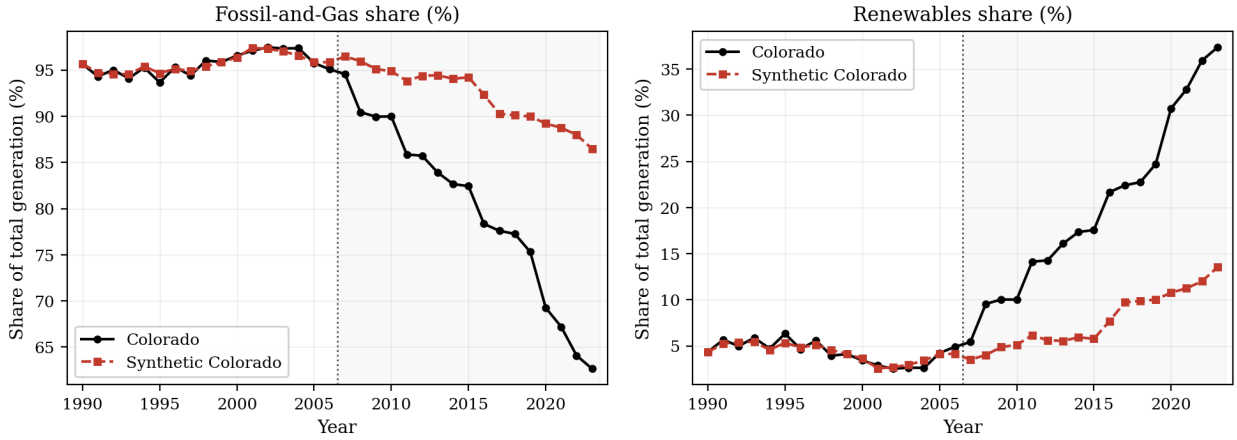

Figure 2. Generation shares by category: Colorado versus synthetic Colorado, 1990–2023. *Note.* The vertical dotted line marks the treatment year (2006).

in absolute terms. The total generation treatment effect (TTE) is near zero throughout, indicating that the renewable expansion came directly at the expense of fossil-fuel generation rather than through a net increase in total output.

Figure 3 presents the share treatment effects year by year. The compositional nature of the effect is transparent: because the two shares sum to one, each gain in the renewable share corresponds to an equal loss in the fossil-and-gas share. The growing magnitude over

Table 2. Estimated treatment effects, 2007–2023

Year	$\mathrm{CTE}_{\mathrm{fos}}$	$\mathrm{CTE}_{\mathrm{ren}}$	TTE	$\mathrm{STE}_{\mathrm{ren}}$
2007	−4.2%	+51.0%	−2.3%	+1.9 pp
2008	−8.5%	+130.1%	−2.9%	+5.5 pp
2009	−10.5%	+94.6%	−5.4%	+5.2 pp
2010	−13.8%	+77.7%	−9.2%	+4.9 pp
2011	−12.1%	+120.1%	−4.0%	+8.0 pp
2012	−14.2%	+139.6%	−5.6%	+8.7 pp
2013	−13.2%	+183.1%	−2.3%	+10.5 pp
2014	−15.1%	+182.0%	−3.4%	+11.4 pp
2015	−21.0%	+174.6%	−9.7%	+11.8 pp
2016	−18.3%	+171.8%	−3.7%	+14.0 pp
2017	−14.2%	+129.9%	−0.2%	+12.7 pp
2018	−16.7%	+123.6%	−2.9%	+12.9 pp
2019	−16.4%	+146.0%	−0.2%	+14.7 pp
2020	−25.8%	+172.4%	−4.5%	+20.0 pp
2021	−25.7%	+186.0%	−1.9%	+21.6 pp
2022	−27.5%	+197.8%	−0.5%	+23.9 pp
2023	−25.0%	+185.9%	+3.5%	+23.8 pp

Note. $\mathrm{CTE}_{k,t} = q^1_{k,1,t}/\hat{q}^0_{k,1,t} - 1$ is the category-level treatment effect. $\mathrm{TTE}_t = S^1_{1,t}/S^0_{1,t} - 1$. $\mathrm{STE}_{\mathrm{ren},t} = \pi^1_{\mathrm{ren},1,t} - \pi^0_{\mathrm{ren},1,t}$ in percentage points. By the unit-sum constraint, $\mathrm{STE}_{\mathrm{fos},t} = -\mathrm{STE}_{\mathrm{ren},t}$ in each year.

time reflects the progressive tightening of the RPS mandate and the declining cost of wind and solar generation.

6.5. **Placebo test.** Figure 4 displays the placebo distribution of post-to-pre RMSPE ratios. I retain the 23 donor states whose pre-treatment RMSPE does not exceed twice that of Colorado ($2 \times 0.081 = 0.162$), following Abadie et al. (2010).

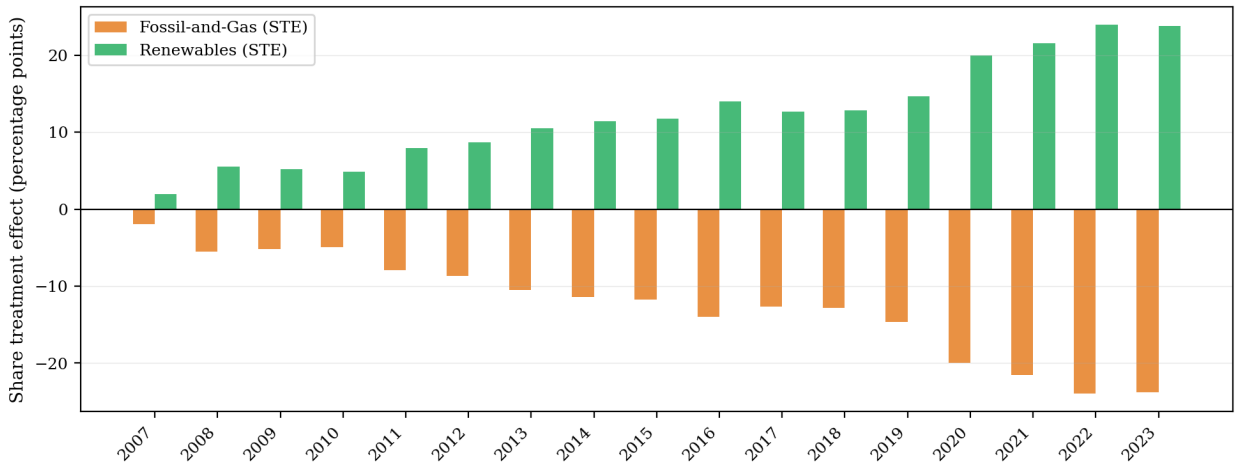

Figure 3. Share treatment effects by category, 2007–2023. *Note.* Each bar shows $\mathrm{STE}_{k,t} = \pi^1_{k,1,t} - \pi^0_{k,1,t}$ in percentage points. The two bars sum to zero in each year by the unit-sum constraint.

Colorado's RMSPE ratio is $R^{(1)} = 8.07$. Among the 23 retained placebo states, Colorado's ratio is the highest, yielding a p-value of 0.0043. This constitutes evidence against the null hypothesis of no treatment effect at the 5% significance level.

The result is noteworthy because the post-2006 period encompasses large common shocks, including the shale gas revolution (2008–2015) and the nationwide wind and solar build-out, which generated large compositional changes in many states. Despite these common trends, Colorado's post-to-pre ratio stands out from the placebo distribution, indicating that the compositional shift in its generation mix was unusually large relative to its pre-treatment trajectory.

7. Conclusion

This paper has proposed compositional synthetic controls, a method for estimating treatment effects when the outcome of interest is a vector of category-specific quantities. The estimator constructs a single synthetic counterfactual vector using one coherent set of weights, obtained by minimizing the squared distance between log-transformed quantity vectors during the pre-treatment period. From this counterfactual vector, treatment effects on individual categories, the aggregate total, and the share composition can all be recovered in a mutually consistent way.

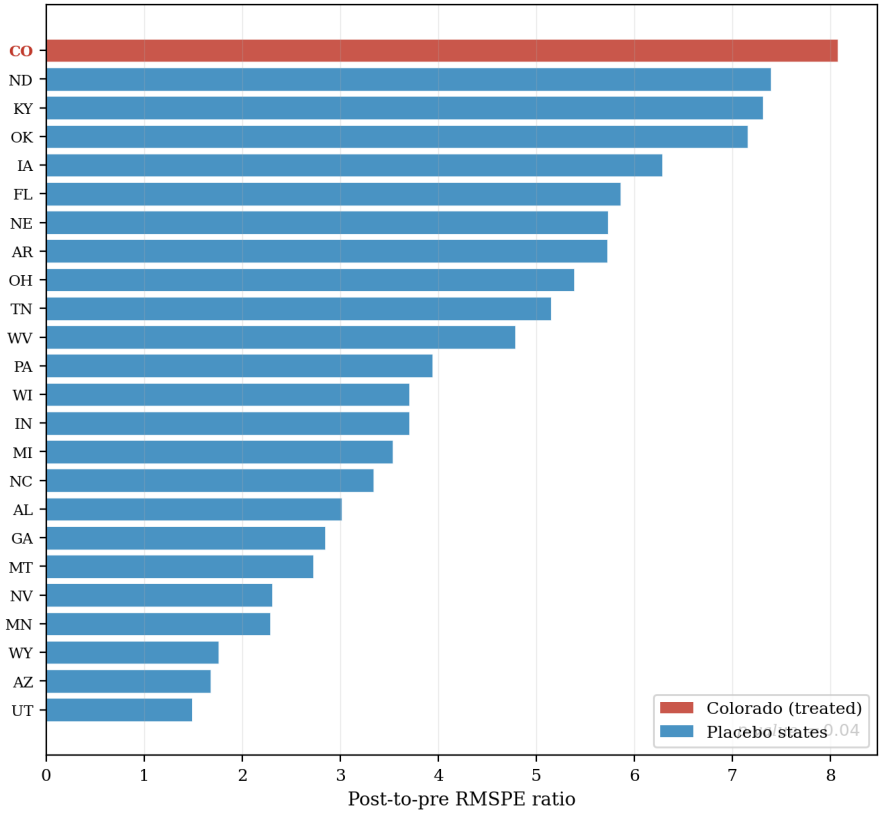

Figure 4. Placebo distribution of post-to-pre RMSPE ratios. *Note.* Colorado is highlighted in red. Only donor states with pre-treatment RMSPE no greater than twice that of Colorado are retained ($N = 23$). Colorado's ratio of 8.07 exceeds all placebo states, yielding a p-value of 0.0043.

Three main results have been established. First, under a random utility model with a shared latent factor structure, the CSC weights admit a behavioral interpretation as measuring preference similarity between the treated unit's representative agent and those of the control units. Second, the synthetic shares produced by CSC coincide with the Fréchet barycenter of the control shares under the Aitchison geometry, the natural metric for compositional data, ensuring geometric coherence with the simplex structure. Third, a placebo-based inference procedure extends the standard synthetic control permutation test to the multivariate compositional setting.

Extensions to settings with multiple treated units, staggered adoption, or partially observed compositions can also be of practical interest for future developments.

8. Acknowledgements

I am grateful to Marc Henry and all the participants of the Marc Henry students' econometrics study group at Penn State for their valuable comments. All remaining errors are my own.

9. Data Availability Statement

The data that support the findings of this study are openly available on: https://www.eia.gov/electricity/

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Appendix A. Proof of Proposition 1

We verify that equations (3) and the expected utility characterization are equivalent under the RUM of Section 4.

From equation (10), for every category k and every unit j:

$$\log q^0_{k,j,t} = \mathbb{E}[U^0_{k,j,t}] - \gamma.$$

The raw counterfactual construction (3) requires that

$$\log q^0_{k,1,t} = \sum_{j=2}^{J} \lambda^*_j \log q_{k,j,t} \quad \text{for each } k.$$

Substituting the relation between log-quantities and expected utilities on both sides:

$$\begin{aligned}
\mathbb{E}[U^0_{k,1,t}] - \gamma &= \sum_{j=2}^{J} \lambda^*_j \big(\mathbb{E}[U^0_{k,j,t}] - \gamma\big) \\
&= \sum_{j=2}^{J} \lambda^*_j \mathbb{E}[U^0_{k,j,t}] - \gamma \sum_{j=2}^{J} \lambda^*_j \\
&= \sum_{j=2}^{J} \lambda^*_j \mathbb{E}[U^0_{k,j,t}] - \gamma,
\end{aligned}$$

where the last step uses $\sum_{j=2}^{J} \lambda^*_j = 1$. Cancelling γ from both sides gives

$$\mathbb{E}[U^0_{k,1,t}] = \sum_{j=2}^{J} \lambda^*_j \mathbb{E}[U^0_{k,j,t}] \quad \text{for each } k.$$

Stacking over all k yields $\mathbb{E}[U^0_{1,t}] = \sum_{j=2}^{J} \lambda^*_j \mathbb{E}[U^0_{j,t}]$, as claimed. □

Derivation of (10). Under the RUM decomposition (6), the latent utility is

$$U^0_{k,j,t} = \mu^0_{k,j,t} - \log\Big(\sum_{\ell} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t} + \varepsilon^0_{k,j,t}.$$

Since $\varepsilon^0_{k,j,t} \sim \text{Gumbel}(0,1)$ with mean γ, the expected utility is

$$\mathbb{E}[U^0_{k,j,t}] = \mu^0_{k,j,t} - \log\Big(\sum_{\ell} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t} + \gamma.$$

From (8), the log-quantity is

$$\log q^0_{k,j,t} = \mu^0_{k,j,t} - \log\Big(\sum_{\ell} e^{\mu^0_{\ell,j,t}}\Big) + \log S^0_{j,t}.$$

Subtracting the second expression from the first yields $\mathbb{E}[U^0_{k,j,t}] = \log q^0_{k,j,t} + \gamma$, i.e., $\log q^0_{k,j,t} = \mathbb{E}[U^0_{k,j,t}] - \gamma$. □

Appendix B. Proof of Proposition 2

We show that the counterfactual shares produced by CSC equal the Fréchet barycenter of the control shares under the Aitchison geometry.

The Fréchet barycenter of a finite collection of points in a simplex equipped with the Aitchison metric has a well-known closed form. Because the Aitchison metric is induced by an isometric embedding of the simplex into $\mathbb{R}^{p-1}$ via the centred log-ratio (or equivalently the log-odds map ℓ), the Fréchet barycenter of $\{\pi_{j,t}\}_{j=2}^J$ with weights $\{\lambda^*_j\}$ is

$$\ell^{-1}\Big(\sum_{j=2}^J \lambda^*_j\, \ell(\pi_{j,t})\Big).$$

It therefore suffices to show that the CSC shares in (4) (in the absence of a smearing correction) satisfy this representation.

Without the smearing correction, the counterfactual quantities are $q^0_{k,1,t} = \exp\big(\sum_j \lambda^*_j \log q_{k,j,t}\big)$. The log-odds ratio of category k relative to the reference category p is

$$\begin{aligned}
\log \frac{\pi^0_{k,1,t}}{\pi^0_{p,1,t}} &= \log \frac{q^0_{k,1,t}}{q^0_{p,1,t}} \\
&= \sum_{j=2}^J \lambda^*_j \log q_{k,j,t} - \sum_{j=2}^J \lambda^*_j \log q_{p,j,t} \\
&= \sum_{j=2}^J \lambda^*_j \log \frac{q_{k,j,t}}{q_{p,j,t}} \\
&= \sum_{j=2}^J \lambda^*_j \log \frac{\pi_{k,j,t}}{\pi_{p,j,t}},
\end{aligned}$$

where the last equality uses the fact that $S_{j,t}$ cancels in the ratio $q_{k,j,t}/q_{p,j,t} = \pi_{k,j,t}/\pi_{p,j,t}$.

This holds for every $k = 1, \ldots, p-1$. In vector form,

$$\ell(\pi^0_{1,t}) = \sum_{j=2}^J \lambda^*_j\, \ell(\pi_{j,t}),$$

and applying ℓ^{-1} to both sides:

$$\pi^0_{1,t} = \ell^{-1}\Big(\sum_{j=2}^{J} \lambda^*_j \, \ell(\pi_{j,t})\Big),$$

which is precisely the Fréchet barycenter characterization. $\square$